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A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Jr.Super60_NUCLEUS BT

2022_P1

Date: 15-10-2023

Time: 02.00Pm to 05.00Pm

CTA-12

Max. Marks:180

MATHEMATICS

: Quadratic Equations, Sequence & Series, Matrices,
Determinants & Linear Equations

CHEMISTRY

: HYDROGEN AND ITS COMPOUNDS, GEOMETRICAL
AND OPTICAL ISOMERISM, GROUP-13, GROUP-14

PHYSICS

: Thermodynamics, Rotation

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2022-P1-M odel**

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Numerical Value Answer Type	+ 3	0	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 15 – 18)	Matching Type	+ 3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Numerical Value Answer Type	+ 3	0	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 33 – 36)	Matching Type	+ 3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+ Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Numerical Value Answer Type	+ 3	0	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+ 4	-2	6	24
Sec – III(Q.N : 51 – 54)	Matching Type	+ 3	-1	4	12
Total				18	60

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Space for rough work

Page 2

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023****JEE MAIN 2023**SINGARAJU
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Bachchan Class
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RANK**RANK 1**



MATHEMATICS

Max Marks: 60

SECTION - I
(NUMERICAL VALUE TYPE)

This section contains EIGHT (08) questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 ONLY if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

01. Let $S = \sum_{k=1}^{\infty} \frac{2^k}{7 \cdot 2^k + 1}$, then the value of $\frac{1}{S}$ is

02. If $A = (49 + 20\sqrt{6})^{1/4}$, $B = 8\sqrt{3} + \frac{8\sqrt{6}}{\sqrt{3}} + \frac{16}{\sqrt{3}} + \frac{16\sqrt{2}}{3} + \dots$ and $|\alpha - \beta| = (6\sqrt{6})^k$, where

α, β are the roots of the quadratic equation $A(\sqrt{3} - \sqrt{2})x^2 + \frac{B}{(\sqrt{3} + \sqrt{2})}x + C = 0$ and

$k = \log_6 10 - 2\log_6 \sqrt{5} + \log_6 \sqrt{(\log_6 18 + \log_6 72)}$, then the value of C is

03. Let $p(x) = 0$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $p(x) = 0$ will be

04. Given $a < b < c$ and a, b, c are natural numbers. Also, a, b, c are in Harmonic Progression and $a = 20$ and b divides c . The number of triplets (a, b, c) satisfying the given conditions are

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VENKAT KOUNDINYA
AIR 100 (2023)
Sri Chaitanya
JEE 12th Class
300
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1JEE Advanced
2023VAVILALA
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Sri Chaitanya
JEE 12th Class
341
360RANK
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AIR 100 (2023)
Sri Chaitanya
JEE 12th Class
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05. Given $x, y, z \in R$, $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = 4$, then the number of positive even divisors of the absolute value of $(x-y)(y-z)(z-x) \begin{vmatrix} yz & zx & xy \\ y+z & z+x & x+y \\ 1 & 1 & 1 \end{vmatrix}$ will be
06. The set of real parameter 'b' for which the equation $x^4 - 4bx^2 + x - 2b + 4b^2 = 0$ has all real solutions is given by $\left[\frac{p}{q}, \infty\right)$ where p and q are natural numbers having their highest common factor as 2, then the value of $(p+q)$ will be
07. If $x, y, z \in R - \{0\}$ such that $\begin{vmatrix} a & b-y & c-z \\ a-x & b & c-z \\ a-x & b-y & c \end{vmatrix} = 0$, then the value of the square of the reciprocal of $\frac{a}{x} + \frac{b}{y} + \frac{c}{z}$ is
08. If $a, b, c \in R - \{0\}$, $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a common root, then the value of $3^{\frac{a^2}{bc}} \cdot 3^{\frac{b^2}{ac}} \cdot 3^{\frac{c^2}{ab}}$ will be

SECTION - II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

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Space for rough work

Page 4

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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
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CHANDRILAS REDDY
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09. Given, $f(x) = \sum_{n=0}^{\infty} \frac{x^{2^n}}{1-x^{2^{n+1}}}$, $0 < x < 1$ and $g(x) = \sum_{n=0}^m \frac{2^n}{x^{2^n} + 1}$, $x \in \mathbb{R}^+ - \{1\}$. Then which of the following is/are **CORRECT**?
- A) $f(x) = \frac{x}{1-x}$ B) $f\left(-\frac{2}{3}\right) = -\frac{2}{5}$
- C) $g(x) = \frac{1}{x-1} - \frac{2^{m+1}}{x^{2^{m+1}} - 1}$ D) $g(0) = 2^{m+1} - 1$
10. If the system of equations in (x, y, z) is as follows: $(a-t)x + by + cz = 0$, $bx + (c-t)y + az = 0$ and $cx + ay + (b-t)z = 0$ has a non-trivial solution, then which of the following is/are TRUE? (for all possible sets of a, b, c)
- A) t can be $(a+b+c)$ B) t^2 can be $(a^2 + b^2 + c^2 - ab - bc - ca)$
- C) $t^2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ D) $t^2 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & a & c \end{vmatrix}$
11. If α, β^2 are the roots of $5x^2 - 20x + 3k = 0$ and β, α^2 are the roots of $kx^2 - 50x + 45 = 0$, where $\alpha, \beta \in \mathbb{R}$ and $k \neq 0$, then which of the following is/are Correct?
- A) $\alpha\beta = 3$ B) $k = 5$ C) $\alpha^3 - 10\alpha + 3\beta = 0$ D) $\beta^3 - 4\beta + \alpha = 0$
12. Which of the following statement(s) is/are **FALSE**?
- A) If A and B are square matrices of the same order such that $ABAB = O$, then it follows that $BABA = O$
- B) Let A and B be different $n \times n$ matrices with real numbers. If $A^3 = B^3$ and $A^2B = B^2A$, then $(A^2 + B^2)$ is invertible
- C) If A is a square, non-singular and symmetric matrix, then $\left((A^{-1})^{-1}\right)^{-1}$ is skew symmetric
- D) The matrix of the product of two invertible square matrices of the same order is also invertible

JEE MAIN
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RANK 1
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2023VAVILALA
CHANDRILAS REDDY
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13. If $P = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ and $P^{2023} = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$, then which of the following is/are correct?
- A) $p = s$ B) $p + q + r + s = 4026$
 C) $p^2 + q^2 + s^2 = 2$ D) $q = 2023$
14. If $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + fn + d, \forall n \geq 2023$, then which of the following is/are correct?
- A) $a + c = b + d$
 B) $d = 0$
 C) $a, b - \frac{2}{3}, c - 1$ are in Arithmetic Progression
 D) $\frac{a}{c}$ is an integer

SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 In all other cases.

15. If $a, b, c \in R, A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is an orthogonal matrix, then match the following:

Column-I	Column-II
(I) $a^2 + b^2 =$	(P) 1
(II) $b^2 - c^2 =$	(Q) $\frac{5}{6}$
(III) $c^2 + a^2 =$	(R) $-\frac{1}{6}$
(IV) $a^2 + b^2 + c^2 =$	(S) $\frac{2}{3}$

JEE MAIN
2023SINGARAJU
VENKAT KUMAR
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2023VAVILALA
CHANDRILAS REDDY
RANK 1
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- A) (I)-(S); (II)-(R); (III)-(Q); (IV)-(P) B) (I)-(S); (II)-(P); (III)-(Q); (IV)-(R)
C) (I)-(Q); (II)-(R); (III)-(S); (IV)-(P) D) (I)-(Q); (II)-(P); (III)-(S); (IV)-(R)

16. Let $S_n = 1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$ 'n' terms. Column-II contains the units digit of the term given in Column-I.

Column-I	Column-II
(I) $\frac{3S_{2024}}{506}$	(P) 6
(II) $\frac{3S_{2023}}{506}$	(Q) 4
(III) $\frac{6S_{2022}}{1011}$	(R) 0
(IV) $\frac{6S_{2021}}{1011}$	(S) 2

- A) (I)-(R); (II)-(R); (III)-(S); (IV)-(P)
 B) (I)-(R); (II)-(Q); (III)-(Q); (IV)-(P)
 C) (I)-(Q); (II)-(R); (III)-(S); (IV)-(R)
 D) (I)-(Q); (II)-(P); (III)-(S); (IV)-(R)

17.

Column-I	Column-II
(I) The number of real solution(s) of the equation $x^2 - x - 2 = 0$ is/are	(P) 0
(II) If one of the roots is the square of the other for the equation $3x^2 + px + 3 = 0, p > 0$ then $ p =$	(Q) 1
(III) The number of real value(s) of k for which the system of equations $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k - 1$ have infinitely many solutions is/are	(R) 2
(IV) Number of real root(s) of $x^2 - \frac{2}{x-2} = 4 - \frac{2}{x-2}$ is/are	(S) 3
	(T) 6

- A) (I)-(S); (II)-(R); (III)-(Q); (IV)-(R) B) (I)-(S); (II)-(P); (III)-(Q); (IV)-(Q)
C) (I)-(R); (II)-(S); (III)-(Q); (IV)-(Q) D) (I)-(R); (II)-(S,T); (III)-(Q); (IV)-(R)



18. Let p, q, r be non-zero real numbers that are respectively the $10^{\text{th}}, 100^{\text{th}}$ and 1000^{th} terms of a harmonic progression. Consider the system of linear equations:
- $$x + y + z = 1, 10x + 100y + 1000z = 0, (qr)x + (pr)y + (pq)z = 0$$

Column-I	Column-II
(I) If $\frac{q}{r} = 10$, then the system of linear equations has	(P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II) If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10$, then the system of linear equations has	(S) no solution
	(T) at least one solution

The correct option is:

- A) (I)-(T); (II)-(R); (III)-(S); (IV)-(T) B) (I)-(Q); (II)-(S); (III)-(S); (IV)-(R)
 C) (I)-(Q); (II)-(R); (III)-(P); (IV)-(R) D) (I)-(T); (II)-(S); (III)-(P); (IV)-(T)

JEE MAIN
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Sri Chaitanya
Bachchan Class
300
300RANK
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2023VAVILALA
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Bachchan Class
341
360RANK
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PHYSICS

Max Marks: 60

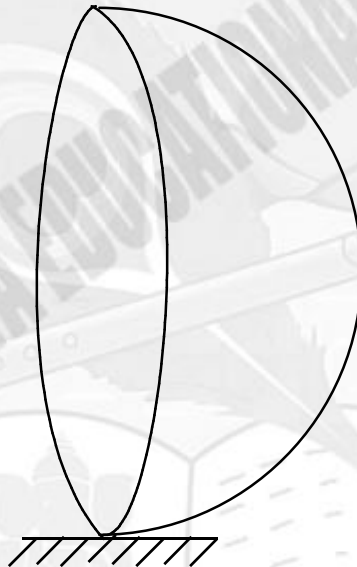
SECTION - I
(NUMERICAL VALUE TYPE)This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

19. A uniform hemispherical shell of radius $R=1\text{m}$ is initially at rest and its orientation is shown in figure (diametric plane is vertical). The horizontal surface is sufficiently rough for no slipping. It is released from this position. When it has turned by 30° , find its angular acceleration in (rad/s^2) (Take $g=9.8\text{m/s}^2$)



20. A vertical cylinder of cross-sectional area $S=10^{-3}\text{m}^2$ contains 1 mol of an ideal monatomic gas under a piston of mass $M=10\text{kg}$. At a certain instant, a heater is switched on which transmits 3500J of heat per second to the gas. If the entire system is thermally insulated and the pressure of the gas under the piston remains constant, then what is the speed (in m/s) of the piston? Take atmospheric pressure to be $P_0=10^5\text{Pa}$.

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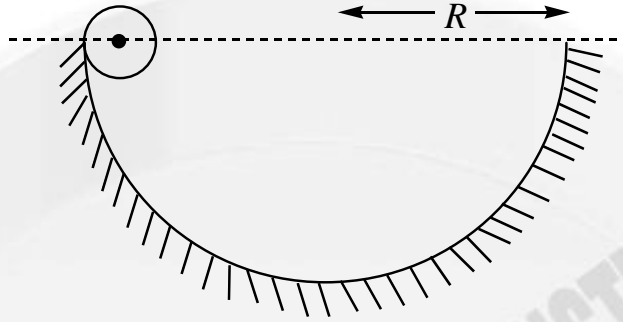
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Page 9

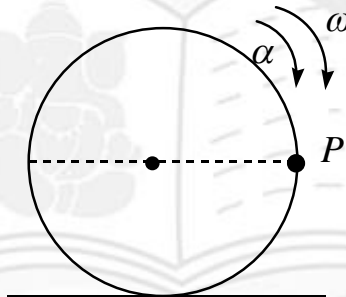
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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023JEE MAIN
2023SINGARAJU
VENKAT KOUNDINYA
AIR 100 (2023)
Sri Chaitanya
Bachchan Class
300
300RANK
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2023VAVILALA
CHIVILAS REDDY
AIR 100 (2023)
Sri Chaitanya
Bachchan Class
341
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1NEET
2023BORA VARUN
CHAKRAVARTY
AIR 100 (2023)
Sri Chaitanya
Bachchan Class
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21. In the figure shown, a small solid ball of mass 'm' can move without sliding in a fixed cylindrical track of radius R in vertical plane. It is released from the top. The resultant force on the ball when it is at the lowest point of the track is given by $k(mg)$, then k is

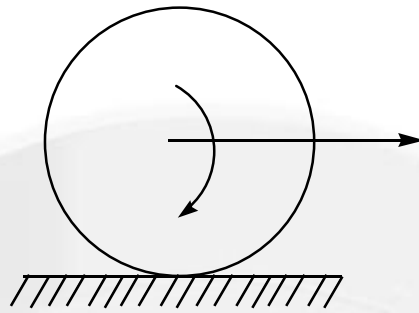


22. A spherical balloon is filled with an ideal mono-atomic gas. At some instant its pressure was P_0 , volume V_0 and temperature T_0 . The balloon expanded in such a manner so that the rate of radiation from the balloon remained constant. It was found that the work done by gas inside the balloon was $6P_0V_0\left[1-\frac{1}{\sqrt{n}}\right]$ when its volume increased to $8V_0$. Find the value of n.
(assume ambient temperature to be close to 0K)
23. A ring of radius $R=1\text{m}$ is rolling on a floor. At an instant its angular velocity is $\omega=1\text{rad/s}$ and angular acceleration is $\alpha=1\text{rad/s}^2$. Point P on the ring is at height $h=1\text{m}$ from ground at this instant as shown in figure. At this instant, find radius of curvature of trajectory of point P (in m)

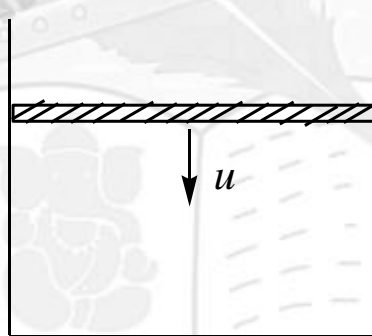


24. A uniform solid sphere of mass M and radius r slips on a rough horizontal plane. At some instant it has velocity of its centre of mass u_0 and angular velocity $\frac{u_0}{2r}$ as shown in the figure. The velocity of its centre of mass when the sphere starts pure rolling motion is found to be nu_0 . Find the value of n.

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AIR NO. 1 (2023)
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300
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25. A certain amount of mono-atomic ideal gas undergoes a process $pu^\eta = C$, where p is the density of the gas and u is the internal energy of the gas. It was found that the ratio $r = \frac{\Delta W}{\Delta Q}$ for the process was $r = 2/3$. What is the value of η ? (ΔW is work done by the gas and ΔQ is heat given to gas)
26. Two moles of an ideal monoatomic gas is enclosed in an adiabatic vertical cylinder fitted with a smooth adiabatic massive piston initially at rest. The whole system is placed inside a highly evacuated chamber. If the piston is imparted an initial velocity $u = 8 \text{ m/s}$ in vertical downward direction, after making a large number of oscillations, the piston finally comes to a complete stop. Find the displacement (in meter) of the piston. (Take $g = 10 \text{ m/s}^2$)



SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

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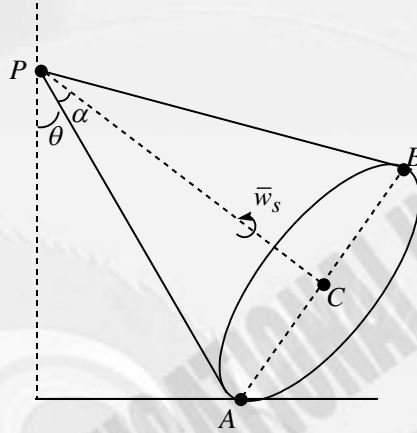
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Page 11

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Sri Chaitanya
Bachchan Class
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CHANDRILAS REDDY
AIR 100 (2023 JEE)
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Bachchan Class
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1NEET
2023BORA VARUN
CHAKRAVARTHI
AIR 100 (2023 JEE)
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27. A right circular cone of base radius R and half apex angle α precesses about a vertical axis passing through its apex P such that its slant length is making angle θ with vertical as shown in the figure. It is spinning about its own axis with angular velocity ω_s . There is no slipping at the point of contact of cone and horizontal surface. At the instant shown A is a point on cone which is in contact with ground and B is diametrically opposite point on its base. C is centre of base circle. At this instant:

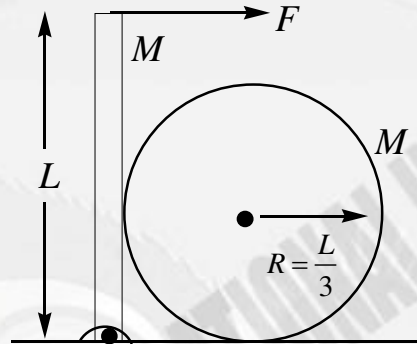


- A) Speed of point C on cone is $\omega_s R \sin(\theta + \alpha) \cdot \frac{\cos \alpha}{\sin \theta}$
- B) Speed of point C on cone is $\frac{\omega_s R \sin(\theta + \alpha)}{\sin \theta}$
- C) Speed of point B on cone is $2\omega_s R \sin(\theta + \alpha) \cdot \frac{\cos \alpha}{\sin \theta}$
- D) Speed of point B on cone is $\frac{2\omega_s R \sin(\theta + \alpha)}{\sin \theta}$
28. Which of the following statement is/are **INCORRECT** about a solid sphere rotating about a fixed axis passing through its center, with constant angular velocity ? Consider $r < R$, where R is radius of sphere.
- A) All points in sphere at perpendicular distance ' r ' from the axis same velocity
- B) The magnitude of acceleration of all points in sphere at perpendicular distance r from the axis is same
- C) No two points on the sphere can have same acceleration
- D) All points on the surface of the sphere have same speed.

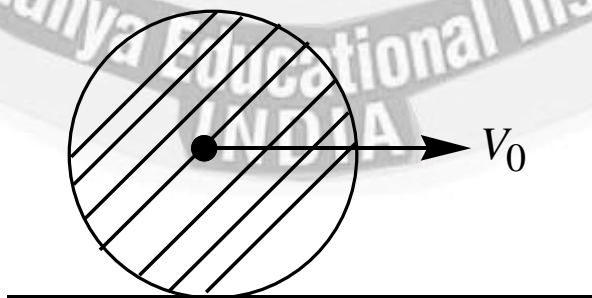




29. A uniform rod of mass M and length L is hinged at lower end on a table. The rod can rotate freely in vertical plane and there is no friction at the hinge. A uniform solid ball of mass M and radius $R = \frac{L}{3}$ is placed in contact with the vertical rod and a horizontal force F is applied at the upper end of the rod. There is no friction between rod and the ball and there is sufficient friction between ball and ground for pure rolling. Mark the correct statement(s).



- A) The acceleration of centre of mass of ball just after the application of force is $\frac{15}{22} \frac{F}{M}$.
- B) The angular acceleration of rod just after the application of force is $\frac{45}{22} \frac{F}{ML}$.
- C) The horizontal component of hinge reaction on the rod just after application of force is $\frac{43}{44} F$.
- D) The horizontal component of hinge reaction on the rod just after application of force is $\frac{44}{43} F$.
30. A uniform disc of mass m and radius R starts with velocity V_0 on rough horizontal floor with a pure translational motion at $t=0$ as shown. At $t=t_0$, disc starts rolling without sliding. Then (Coefficient of frictions is μ)

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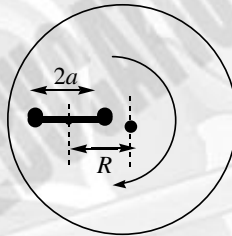
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- A) Net work done by frictional force upto time $t \leq t_0$ is given by $\frac{m\mu g t}{2}(3\mu g t - 2V_0)$
- B) Net work done by frictional force upto time $t \leq t_0$ is given by $\frac{m\mu g t}{2}(2\mu g t - V_0)$
- C) Net work done by frictional force upto time $t=2t_0$ is given by $m\mu g t_0(2\mu g t_0 - V_0)$
- D) Net work done by frictional force upto time $t=2t_0$ is given by $\frac{m\mu g t_0}{2}(3\mu g t_0 - 2V_0)$

31. A dumbbell consists of two particles of equal mass m (connected with massless rod) separated by a distance $2a$. It is initially rigidly attached along a radial line to a disk that is rotating at angular velocity ω rad/s about a fixed vertical axis (see figure). The center of the dumbbell is at a distance R from the center of the disk. Choose the CORRECT statement(s) about the motion of dumbbell after it is suddenly free from disk. Assume no friction between dumbbell and disk and all the motion occurs on a horizontal frictionless plane.



- A) Motion of dumbbell will be pure translational.
- B) Angular velocity of dumbbell remains same i.e. ω .
- C) Kinetic energy of dumbbell will be given by $\frac{1}{2}mR^2\omega^2$.
- D) Kinetic energy of dumbbell will be given by $ma^2\omega^2$.
32. An ideal gas has a molar specific heat capacity C_v at constant volume. The molar specific heat capacity of this gas as a function of its volume V
- A) is $\left(C_v + \frac{R}{\alpha V}\right)$ if the gas undergoes the process $T=T_0e^{\alpha V}$
- B) is $\left(C_v + \frac{2R}{\alpha V}\right)$ if the gas undergoes the process $T=T_0e^{\alpha V}$
- C) is $\left\{C_v + \frac{R}{(1+\alpha V)}\right\}$ if the gas undergoes the process $P=P_0e^{\alpha V}$
- D) is $\left(C_v + \frac{R}{\alpha V}\right)$ if the gas undergoes process $P=P_0e^{\alpha V}$.





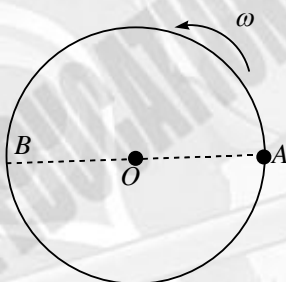
SECTION – III

(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases.

33. A horizontal circular turn table has a boy standing at the edge at point A as shown in the figure. The turn table can rotate without friction about a vertical axis passing through its centre O. It is set into motion with angular velocity ω . Now the boy starts walking slowly with respect to the table along the diameter AB and after some time reaches B. During this time



Column I		Column II	
A)	Moment of inertia of table + boy about a vertical axis through O	P)	Remains constant
B)	The angular velocity of the table.	Q)	First increases, then decreases
C)	The angular momentum of the (table + boy) system about O	R)	First decreases, then increases
D)	Kinetic energy of (table + boy) system	S)	Decreases continuously
		T)	Increases continuously

A) A-R, B-Q, C-P, D-T

B) A-R, B-T, C-P, D-Q

C) A-R, B-Q, C-P, D-Q

D) A-R, B-P, C-P, D-Q



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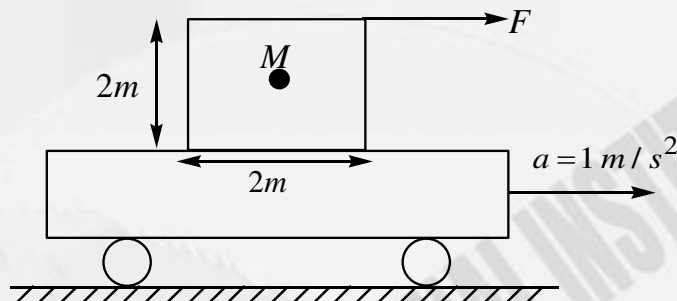
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34. A cubical block having square cross-section of side 2 m and of mass $M=10\text{ kg}$ is resting over a platform moving at constant acceleration $a = 1\text{ m/s}^2$. Coefficient of friction between the block and the platform is $\mu = 0.1$. A horizontal force F acts at the top of the cube as shown in figure. Now match column-I with column-II.
(Take $g = 10\text{ m/s}^2$)



Column-I		Column-II	
A)	$F=0$	P)	Block neither topples nor slips over the platform
B)	$F=85\text{ N}$	Q)	Block topples but does not slip over the platform
C)	$F=15\text{ N}$	R)	Block slips but does not topple over the platform
D)	$F=25\text{ N}$	S)	Block slips as well as topples on the platform

A) A-P, B-Q, C-P, D-R

B) A-P, B-S, C-P, D-R

C) A-P, B-S, C-P, D-Q

D) A-P, B-R, C-P, D-R

35. Column-I shows some bodies (ring, disc, hollow sphere, solid sphere) rolling without slipping on horizontal surface. Two forces F_1 and F_2 are acting at different points parallel to surface. Column-II gives direction and magnitude of static friction between bodies and surface. Match column-II with column-I.

Column-I		Column-II	
A)	Ring	P)	Direction of friction must be in positive x-axis

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B)		Q)	Direction of friction must be in negative x-axis
C)		R)	Magnitude of friction must be 2N
D)		S)	Magnitude of friction must be 4N
		T)	Direction and magnitude of friction can be found if and only if mass and radius is given

A) A-QR, B-QR, C-PS, D-R

B) A-QR, B-QS, C-QR, D-R

C) A-PR, B-QS, C-QS, D-Q

D) A-QR, B-QR, C-QS, D-QR

36. Match the following

Column-I		Column-II	
A)	An ideal gas undergoes a process from A to B	P)	ΔQ can be +ve

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B)	An ideal gas undergoes a process from A to B	Q)	ΔQ can be -ve
C)	An ideal gas undergoes a process from A to B	R)	ΔU can be +ve
D)	An ideal gas undergoes a cyclic process ABA	S)	ΔU can be -ve
		T)	W can be +ve

- A) A-P, Q, R, S, T; B-P, R, T; C-P, R, T; D-P, T
 B) A-P, Q, R, S, T; B-P, S, T; C-P, R, T; D-Q, T
 C) A-P, Q, T, R, S; B-P, Q, T; C-P, R, T; D-S, T
 D) A-P, S, T, Q, R; B-P, T, R; C-P, R, T; D-R, T





CHEMISTRY

Max Marks: 60

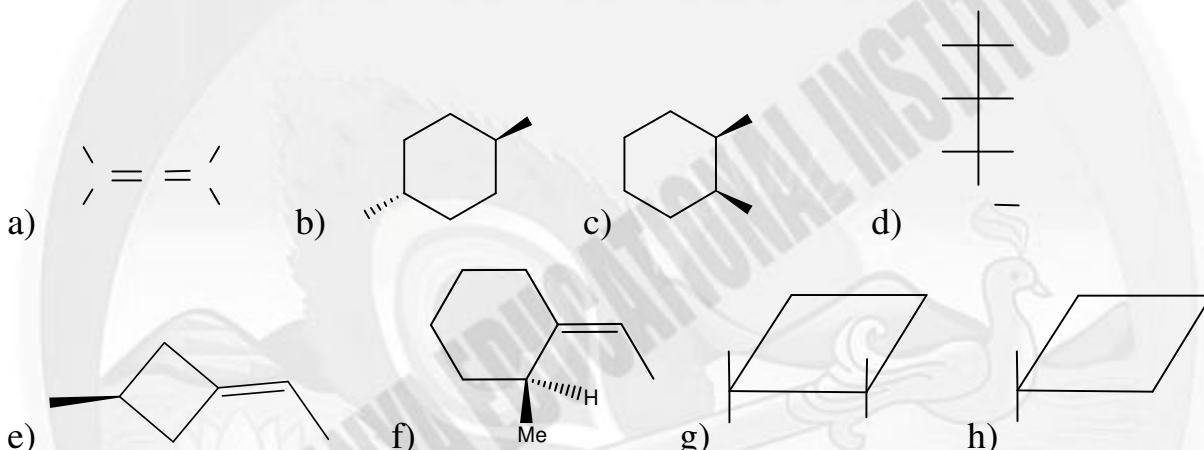
SECTION - I
(NUMERICAL VALUE TYPE)This section contains **EIGHT (08)** questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

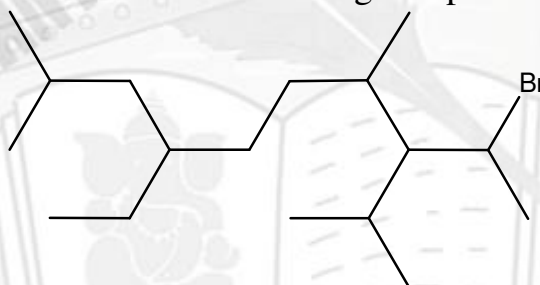
Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

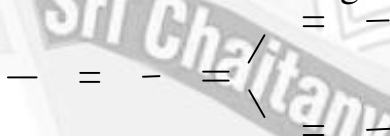
37. How many following compounds are optically active?



38. The total number of stereoisomers for following compounds are



39. Find the total number of geometrical isomerism



40. Amount (in grams) of gaseous product obtained when 2.8 grams of diborane undergoes hydrolysis is: (Given that: GAW of B = 11, H=1, O=16)

41. Fe^{2+} is converted to Fe^{3+} with H_2O_2 in presence of both acidic and basic medium. The ratio (simplest) between number of moles of H_2O_2 required to convert 1 mole of Fe^{2+} to Fe^{3+} in acidic and basic medium respectively is $x : y$, then $(x + y)$ is:

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Page 19

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42. Orthoclase is a crystal in which Al^{+3} substitutes for one fourth of the Si^{+4} ions in SiO_2 . K^+ ions balance the charge. What is the formula weight of orthoclase ?
($AW : Al = 27, Si = 28, K = 39, O = 16, H = 1$)
43. A convenient laboratory method for the preparation of a hydride of boron involves the oxidation of sodium borohydride with iodine. Total moles of gaseous products obtained during the oxidation of 2 moles of sodium borohydride with 1 mole of I_2 is:
44. If the formula of a silicate ion with five silicate units shared at corners in a linear chain is $Si_5O_x^{y-}$, then the value of $\frac{x}{y}$ is:

SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;

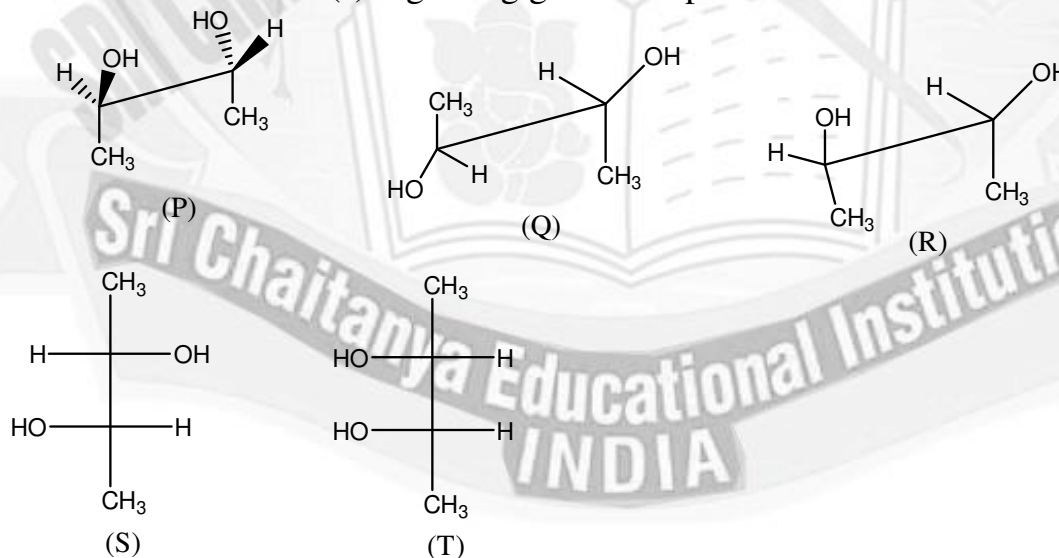
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

45. The correct statement(s) regarding given compounds is/are

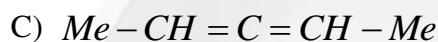
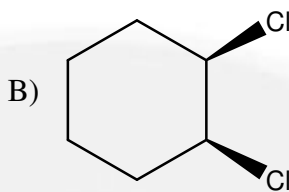
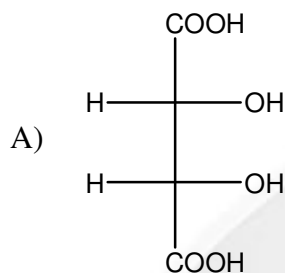


- A) P and Q have different boiling points
B) Q and R have different boiling points
C) P and S have different boiling points
D) Q, R and T are the same compound

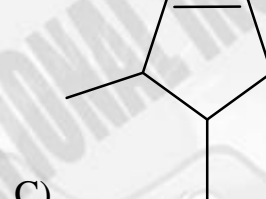
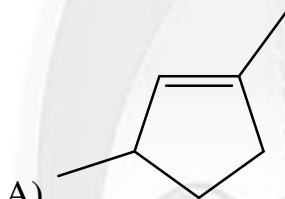
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46. Which of the following compound/s can show optical isomerism?



47. The compounds which cannot show geometrical isomerism are



48. Which of the following order(s) is/are correct?

A) $BCl_3 > AlCl_3$: Lewis acidic nature

B) $BF_3 < BCl_3 < BBr_3$: Lewis acidic nature

C) $SiF_4 > SiCl_4$: Ease of hydrolysis under normal conditions

D) Diamond > Graphite: Thermal conductivity

49. Which of the following is incorrect regarding H_2, D_2, H_2O, D_2O and H_2O_2 ?

A) $H_2 < D_2$: Bond dissociation enthalpy

B) $H_2O < D_2O < H_2O_2$: Dielectric constant

C) $H_2O_2 > H_2O > D_2O$: Boiling point

D) $H_2O_2 > D_2O > H_2O$: Density

50. In which of the following reactions dihydrogen is obtained as one of the product?



SECTION - III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).

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• **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

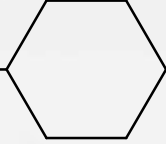
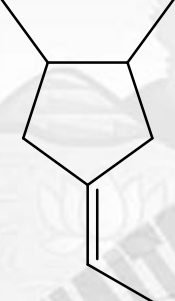
• Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 In all other cases.

51.

Column-I (Compounds)		Column-II (Total number of stereoisomers)	
I)	$\text{CH}_3\text{-CH=CH-}$  -CH=CH-CH_3	P)	10
II)	$\text{CH}_3\text{-CH=CH-}$  -CH=CH-CH_3	Q)	6
III)		R)	2
IV)	$\text{CH}_3\text{-CH=}$  -CH-CH_3	S)	4
		T)	8

A) I – Q; II – P; III – S; IV – R

B) I – Q; II – T; III – S; IV – R

C) I – Q; II – P; III – R; IV – S

D) I – P; II – Q; III – S; IV – R

52.

Column-I (Compounds)		Column-II (Total number of stereoisomers)	
I)	$\text{CH}_3\text{-CH-CH=CH-CH-CH}_3$ Cl Cl	P)	4
II)	$\text{CH}_3\text{-CH=CH-CH-CH=CH-CH}_3$ Cl	Q)	5

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III)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cl}-\text{HC}-\text{Cl} \\ \\ \text{CH}_3-\text{CH}-\text{C}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{Cl} \quad \text{CH}-\text{Cl} \\ \\ \text{CH}_3 \end{array} $	R)	6
IV)	$ \begin{array}{ccc} \text{Cl} & & \text{Cl} \\ & & \\ \text{CH}_3-\text{CH} & & \text{CH}-\text{CH}_3 \\ & \diagdown \quad \diagup & \\ & \text{C}=\text{C} & \\ & \diagup \quad \diagdown & \\ \text{CH}_3-\text{CH} & & \text{CH}-\text{CH}_3 \\ & & \\ \text{Cl} & & \text{Cl} \end{array} $	S)	7
		T	8

A) I – P; II – R; III – Q; IV – S

B) I – R; II – P; III – T; IV – S

C) I – P; II – R; III – T; IV – S

D) I – R; II – P; III – Q; IV – S

53. Balance the reactions given in Column I and identify the products given in Column II for each balanced reaction:

Column I	Column II
I) $\text{MnO}_4^- + \text{H}_2\text{O}_2 \xrightarrow{\text{H}^+}$	P) O_2
II) $\text{MnO}_4^- + \text{H}_2\text{O}_2 \xrightarrow{\text{OH}^-}$	Q) H_2O
III) $\text{PbS} + \text{H}_2\text{O}_2 \xrightarrow{\text{H}^+}$	R) Mn^{4+}
IV) $\text{Mn}^{2+} + \text{H}_2\text{O}_2 \xrightarrow{\text{OH}^-}$	S) MnO_2
	T) Mn^{2+}

Then correct match is:

A) I – PQ; II – PS; III – P; IV – P

B) I – QT; II – PQ; III – P; IV – Q

C) I – PQT; II – PQS; III – Q; IV – R

D) I – PT; II – QS; III – Q; IV – PQ

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54. Match the different types of silicates given in column I with the examples given in column II:

Column I	Column II
I) Ortho silicate	P) Tremolite: $\text{Ca}_2\text{Mg}_2[(\text{Si}_4\text{O}_{11})_2](\text{OH})_2$
II) Pyro silicate	Q) Phenacite: $\text{Be}_2[\text{SiO}_4]$
III) Cyclic silicate	R) Thortveitite: $\text{Sc}_2\text{Si}_2\text{O}_7$
IV) Double chain silicate	S) Beryl : $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$
	T) Crocidolite $\text{Na}_2\text{Fe}_3^{\text{II}}\text{Fe}_2^{\text{III}}[(\text{Si}_4\text{O}_{11})_2](\text{OH})_2$

- A) I – T; II – P; III – S; IV – Q B) I – Q; II – R; III – S; IV – T
 C) I – Q; II – TP; III – S; IV – R D) I – P; II – R; III – R; IV – S



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32 TOP RANKS BELOW 100 IN ALL-INDIA OPEN CATEGORY

RANK 12 DASARI SAKETH NAIDU HT.No. 236060148	RANK 14 GUTHIKONDA ABHIRAM HT.No. 236128121	RANK 17 P GNANA KOUSIK REDDY HT.No. 236135520	RANK 18 D VENKATA YUGESH HT.No. 236048359	RANK 19 MOULIK JINDAL HT.No. 237021004	RANK 22 D PRATAP SINGH HT.No. 232006139	RANK 23 SHAIK SAHIL CHANDA HT.No. 236132870	RANK 26 SONI MAYANK HT.No. 232097025	RANK 33 GOYAL NAMAN HT.No. 222029345
RANK 37 SUMEDH SS HT.No. 237029086	RANK 38 V DEERAJ SATHVIK REDDY HT.No. 236057161	RANK 44 V VERGIYA KAUSHAL HT.No. 237004058	RANK 46 SUTHAR HARSHAL HT.No. 237005060	RANK 58 ABHI JAIN HT.No. 232091036	RANK 61 RIDAM JAIN HT.No. 232068021	RANK 62 SUTHAR HARSH HT.No. 232100028	RANK 63 SOJITRA DIPEN HT.No. 231017001	RANK 65 D PANEENDRANADHA REDDY HT.No. 236050288
RANK 67 GUNDA SUSHANTH HT.No. 236129732	RANK 68 HARSH TAYA HT.No. 237021078	RANK 69 KRISH GUPTA HT.No. 222068018	RANK 73 KUNCHAKURTHY VIVEK HT.No. 236134093	RANK 81 N DHARMA TEJA REDDY HT.No. 236050138	RANK 84 S VENKAT KOUNDINYA HT.No. 236130678	RANK 87 MRUNAL SHRIKANT HT.No. 231104138	RANK 92 SAMOTA APURVA HT.No. 2320999210	RANK 97 N PRINCE BRANHAM REDDY HT.No. 236044112

BELOW 20
All India Open
Category Ranks

10
RANKS (50%)

BELOW 100
All India Open
Category Ranks

32

BELOW 1000
All India Open
Category Ranks

181

BELOW 100
All India
Category Ranks
Count

89

BELOW 1000
All India
Category Ranks
Count

699

NUMBER OF RANKS QUALIFIED
3,621*

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